

ActInf Livestream #030.0 ~ “How to count biological minds: symbiosis, the free energy principle...”

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it's life okay it is hello and welcome everyone thank you to joining the active inference lab it is october 1st 2021 and i'm daniel i'm here with blue and we're going to be talking about this paper how to count biological minds symbiosis the free energy principle and reciprocal multi-scale integration welcome to the active inference lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this page this is recorded in an archived live stream so please provide us with feedback so that we can be improving our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams we are heading into october and the number 30 paper about biological minds we currently have the author matt sims scheduled to come on the october 5th discussion and then we'll have the 30.2 another participatory group discussion so check out the calendar here as well as these other series if you want to join or suggest somebody who we can invite today in acting stream number 30.0 the goal is to learn and discuss this paper how to count biological minds and it's by matthew sims it's from 2020 in the journal synthese and just like all the other zero videos this is just an introduction to some of the ideas some background that blue and i whip together literally leading up to the time of now so it's just an introduction and if you want to go deeper into an idea then come to the discussion we're going to go over the aims and claims abstract roadmap of the paper and then the figures and so on so let's just get started with the introduction readings blue and what makes you excited about this paper you know daniel i'm always interested in studying intelligence and how that scales from the subcellular to the systemic ecosystem level um and so this is kind of an interesting take and it echoes a lot of what we talked about in the discussion of mike levin's paper the computational boundary of the self back in july so i was excited that matthew agreed to come on the live stream and i've been excited about this paper um just because it takes active inference the multi-scale aspect of active in a really enhanced integrative direction how about you what did you like i liked that it combined uh some general

ideas and questions in philosophy philosophy of biology and then tied it to the natural history of a specific species so it's all good you know make claims about distributed systems or the microbiome and the host in general but then tying it to a system that can be studied and has a cool ecological history is a good approach kind of grounds the paper here's the paper and how to count biological minds symbiosis the free energy principle and reciprocal multi-scale integration i'll read the first aim and then blue you can say something or read the other ones to support the notion of symbiotic minds and the notion of nested biological cognizers that falls out of it this paper will develop and deploy the notion of reciprocal multiscale integration from within the free energy principle multiscale integration describes the case where each of two or more cognizing systems uses the other to provide evidence for its own model of the world and itself acting in that world as a case study in support of the notion of symbiotic minds i will look at the symbiosis of vibrio fischeri bacteria and the bobtail squid i will argue that the vibrio squid assemblage constitutes a functionally integrated cognitive hole the markov blanket of which constrains those of vibrio and the squid and in showing how the notion of a symbiotic mind can be supported by within the free energy principle this paper contributes to the philosophy of cognitive science demonstrating that our folk psychological conception of what cogniza what cognizers are require rethinking moreover if the account of symbiotic cognizance presented as tenable by bringing into relief and accounted for at least one possible transition in cognition this paper contributes to a more complete understanding of the evolution of cognition so cool claims and the only kind of two halves to it of the general argument and the specific and symbiosis and ecological association have been approached in the free energy principle framework but i think this reciprocal component it's approached a little differently here than probably in some works but we'll get into that i'll do the first abstract one the notion of a physiological individual has been developed and applied in the philosophy of biology to understand symbiosis an understanding of which is key to theorizing about the major transitions in evolution from multi-organismality to multicellularity the paper begins by asking what such symbiotic individuals can help to reveal about a possible transition in the it makes me think about the transition from a solitary insect cognition to a colony so there's the change in the level of organization and then is that purely cognitive do cognitive changes preceded do they follow what is the relationship between the evolutionary the physiological and the cognitive changes is what the first part is getting it the second part such a transition marks the movement from cooperating individual biological cognizers to functionally integrated cognizing unit somewhere along the way did such cognizant units

simultaneously have cognizers as parts expanding upon the multi-scale integration view of the free energy principle this paper develops an account of reciprocal integration demonstrating how some coupled biological cognizing systems when certain constraints are met can result in a cognizant unit that is in ways greater than the sum of its cognizing parts um and yeah this paper just shouts out a lot to seth mary or smith mary a kid says last name and nader smith and you know the major transitions in evolution uh which is if you haven't read that book or thought about those transitions i mean you can even probably pick up a summary um but it's really um amazing to think about the evolution in terms of the transition from single cell to multi-cell and you know from asexual to sexual reproduction and there's you know i think seven major transitions that happened in the course of evolution and i think that this kind of is hinting at a cognizing evolution or cognitive evolution which we'll we'll get into a little bit later hey parker symbiosis between bee fishery bacteria and the bobtail squid is used to provide an illustration of this account a novel manner of conceptualizing biological cognizers as gradients is then suggested lastly it is argued that the reason why the notion of ontologically nested cognizers may be unintuitive stems from the fact that our folk psychology notion of what a cognizer is has been deeply influenced by our folk biological manner of understanding biological individuals as units of reproduction so that's a really cool point because it involves um our understanding of biology models of biology influence the way that we see the world which is kind of sounds by definition but it's really interesting because people have thought so many different things about different parts of biology about animal biology all of evolution all of the things we see so then how that intersects with our understanding of cognition and what cognitive levels exist what levels of individuality exist cool topics so just a thought here so what uh a biological individual like a what a cognizer is has been in uh influenced by our understanding of reproduction so is a cognizing unit a unit of cognition it's like is there a corresponding in to cognitive evolution as um corresponding units as there are to biological evolution sorry good so those are the main pieces the roadmap not that many sub sections starts with an introduction and goes through a nice description of the free energy principle which we're going to use in multiple slides here there's a distinction drawn which we'll also go into about mirror versus adaptive active inference multi-scale integration will be discussed and different kinds of multi-scale integration come up later on but then there's two sections about this specific bacterial squid symbiosis and then that symbiosis is returned to with this enhanced background on the uni-directional and the reciprocal so that's kind of the overall roadmap keywords or anything to add on that blue all right keywords were free energy

principle active inference symbiosis nested markov blankets multi-scale integration cognitive evolution emergence and physiological individuals so let's start with symbiosis the quote from the paper was symbioses are cooperative heterospecific associations in which each symbiont partner mutually benefits from eg gaining nourishment shelter etc from the presence of the other partners so heterospecific means different species and uh this is a graphic and also those interesting the footnote that just said we're kind of talking about the positive mutualistic form of co-living not commensalist or parasitic forms of co-living or co-existing but even just looking at pairwise interactions before more complex loops are taken into account there's a whole game theory space of interactions what should our models include cooperative interactions adversarial interactions uh which which one of these edges or relationships have what game theory what cognitive theories so it's a it's a cool approach to take on symbiosis that respects all the ways that different systems like a lycan versus a eukaryotic cell with a mitochondria all these different styles so just a thought on that um you know all of these co-living styles are transactional whether it's you know through a voluntary transaction or just straight theft um these kind of associations remind me of how we're we're starting to scale into studying cities and states and thinking about debt right so like the debt that we have to each other like i borrow this or or you borrow that or i benefit you know you in this way and you benefit me in this way and that scales from the you know very basic biological unit all the way through ecosystems so okay physiological individuals so we're gonna come back to this idea of what's an individual but one wrote in is this quote so one way philosophers have addressed this question of individuality is by conceptually developing and applying the notion of a physiological individual a biological individual of this kind is a highly integrated functional unit the heterogeneous parts of which cohere together through regulatory processes like metabolic immunotolerance etc so as to maintain the system's integrity and resist environmental forces of decay and the citation to prado 2011 looks at the role of the immune system and the interaction with the foreign with the other during development and uses that relationship to explore the the notion of the interface between the organism and the environment and looks at like the physiological story involving the microbe and the social niche and all these other components can come into play and that that is in some cases a better explanation or prediction or description of a certain scenario such as cool another cool topic what do you think blue so this this is cool the end of this quote i reject the idea that there is no possible distinction between the organism and its environment i just want to point that out because we'll come back to that in uh in two slides actually yes it's a qualification of the

interface and the system but it's not the rejection that we just give up so it's towards useful partitionings of the system two models of individual individuality more generally there was a quote in the paper that said the theoretical apparatus that falls out of the free energy principle may be used to arrive at an account of the kind of enrichment of living processes that are required for those processes to qualify as cognitive processes ascribable to physiological individuals so that's linking the cognitive perspective with the physiological perspective of individuality making good on the project of strong life-mind continuity so what models of individuality exist we have physiological and cognitive we have the um view that is going to be implicitly described as like folk biological which is the darwinian replicator evolutionary just a few other ones that exist there's like the teleological concept of which is um the end directedness and the cons definition of the organism as an end in it in and of itself and that's when things are acting at the organismal level but there's probably a lot of distance between what was that simplification and what was written the information theory of individuality which i know blue you have something to add on and then integrated information theory as well as pluralism uh and a sort of historical perspective that looks at shifting boundaries of the conception of individuality and then tries to find some organization that doesn't even use that as a central divider i mean these are just some of the ways but just also cool like you know species but then there's so many species concepts but individual seems like that's the one that you can pick out but then it's not always that way well and i guess we'll come back to remark when we uh talk about markov blankets here in a little bit um so i absolutely love love love love this paper um because for many reasons but but i just wanted to highlight um in this paper they talk about how individuality can be nested and it requires a bi-directional information flow which which i think that this distinction um is important right so there has to be like control from the top down and from the bottom up so information has to go both ways and that is what describes an individual so you know if you have um i'm trying to think of like a non-individual like like okay you can have someone broadcasting instructions through a megaphone you know right like like shouting you know like please move to the right please move to the right like so that that's like top down control but there's no feedback from the crowd like the right is blocked there's a path here or there's you know a gap here we can't walk through it or whatever if there's no you know feedback then that's not an individual right so you have this you know top down control but nothing comes back so in an individual there's always that that reciprocal uh flow and then the the forms of individuality that they highlight in this paper are an organism which is like you and me uh a colonial form of uh individual

which is like a biofilm right and then a driven form and this is the part that goes back to that quote about i reject the idea that you know there can be no step or that an organism is not distinct from its environment and so in this enviro this is environmentally driven um and that's this last form of individuality that they highlight so it's aware like the nature of the individual is entirely dependent upon the environment and allowing for this type of individual to exist really allows the formation of life in the very beginning because if you just have collections of molecules you know without a boundary or a barrier that's environmentally driven and so without that uh organism being able to arise from its environment you don't get the origin of life and so i just wanted to just highlight that nice great points the next key word is the free energy principle or fep and this is from active livestream 14 from the illusion that carl firsten made to the wood lice and his kind of observations of nature leading eventually an integrative approach that includes some of these areas like considering how multi-scale systems work and how collectives work so how things work laterally within a level like across and estimates and then multiscale like the colony inside of the ecosystem information and thermodynamics and control theory and action selection as a type of inference we're going to hear it here from the author and look through the four main points that they drew out of the fep so this is again coming from the sims 2020 paper so read for more detail the free energy principle starts from a particular view of life that is grounded in statistical physics so that's an introductory claim an interesting way to start and again just pulling a few highlights out to to get at this first key point a biological system is one that self organizes to a limited set of attracting states that is far from thermodynamic equilibrium so there it's like our body temperature is kept above the room temperature far from the sort of naive chemical equilibrium and that's being maintained despite dissipation being a prevalent in the universe and that resistance to falling to equilibrium is called the non-equilibrium steady state density importantly the nest density towards which an organism's dynamics flow corresponds to its phenotype i.e its regular patterns of behavior morphology and physiology to find itself in its characteristic or phenotypic states provides the organism evidence that its behavior is countering dispersive effects of random fluctuations even static seeming phenotypes like the length of your femur it's actively maintained by cells being renewed and the turnover rate and so on so sometimes it's easier to think about some phenotypes like cognitive phenotypes as actually embodying parametric values that are being kept away from equilibrium other times for more static components of phenotype it's not that way but it's just a cool way to think about phenotype as just these measurements that are stable over some time frame about the world and the

persistence of those measurements through time represents the implementation of some strategy that resisted dissipation hence the maintenance of a non-equilibrium steady state through some kind of strategy anything that blue no the second the right the this brings us to the second primary feature of fep the markov blanket formalism this is used in the context of fep to describe a particular kind of statistical organization of a system relative to that which the system is not namely it describes the statistical partitioning of internal states and then using just the variable names as they are here internal states are the external states ϕ and the sensory states s active states a these latter states being in the markov blanket which are the sensory and the active states so what what about this blue just i mean we've talked um well on the live stream a lot about markov blankets and how they can be used to describe um or or individual individuals so really or to delineate the boundaries of an individual and technically the the markup blanket defines conditional independence so like under what settings or under what settings are these two systems conditionally independent yes so there's some interface that is being used and there's a conditional independence across either side but yeah cool a topic that's come up many times and i'm sure this is not the end of it this brings us to a third essential feature of fep the notion of a generative model a generative model is a probabilistic model that describes how the evolution of sensory states of a markov blanket could be caused by external states it captures prior beliefs in the form of probability distributions about unobserved external states and a likelihood mapping external states to the evolution of sensory states such models are quote implicit in the dynamics of internal states palacios at all 2020. the mapping between external states and the evolution of estimates of internal states is this a matrix that is connecting the observations being made through time to the s matrix so that's a bi-directional relationship with the generative and the recognition model that's the tail of two densities but that's the signal processing uh sort of uh sampling the external world and then having an ongoing estimate of world states which could be just unobserved world states or causes in the world then there's how those states change their time b how policy interacts with how those states change through time π d the prior over the states and then the decision making of π with the c g and the e so this is the partially observable markov decision process the pomdp framing and it's not the only way we can look at it but this gets like many of the key variables down which is why we and there's uncertainties on all of these as we've found so this just gets a lot of the key pieces down the generative model could have a different structure though this is not the only structure so here is from the paper importantly uh you want to read this one sure importantly a systems generative model may be cast in terms of its non-

equilibrium steady state to see how this is the case requires understanding the notion of dual information geometry of self-organizing systems and he gives several citations one of which i guess is this one yep so this is just giving the first uh not like first in literature just like a first introductory description on this dual information geometry this dual aspect concerns the intrinsic information geometry of the probabilistic evolution of internal states so that's like the actual if there's a million neurons and they can each take 10 states that's this the geometry of the internal states is like that many neurons can take on that many values that's the internal true state space of the actual internal and the separate in extrinsic information geometry of probabilistic beliefs about external states that are parameterized by internal states so that n neurons could be fitting a binary model is the light switch up or or it could be a three-state model is it up down or not either or both whatever it's specifying that is a lower dimensionality than the actual state space so it's like there's a lot of decisions in the whole computer but then there's this external realized component that's a smaller state space the extrinsic information geometry we call these intrinsic ie mechanical or state-based and extrinsic i.e markovian or belief-based information geometries and uh wanja has come on just to discuss several papers so definitely look at the work by all of these authors because it's like interesting topic we could probably learn more about that and i know there's people who are listening who know more about it here is those intrinsic and extrinsic information geometries being used in i think the sims paper but uh not 100 sure actually but we can see either that or it's in this previous fristen paper so there's the probabilistic flow the system states over time that's the intrinsic geometry and that's that yeah what i think this is the first paper okay okay and here's the extras so then basically this is just a visual showing the higher dimensionality of the actual components of the system that can project onto like just two lower areas the fourth part of fep uh in the sims paper this brings us to the fourth primary feature of fep active inference i'll read the yellow and then you can give a thought on where you think acting fits in fep proposes that living systems are able to resist the second law of thermodynamics in virtue of avoiding sensory states which are deleterious and actively bringing about those sensory states which allow them to maintain their structural and functional integrity it is by engaging in this coupled process of active inference that an organism minimizes not only current free energy encountered but expected free energy i.e the free energy that would arise were a particular action policy selected and followed um so instead of giving a thought i'm going to give an author's thought on this just to read a quote from the paper the reason for using the free energy principle to investigate and argue for a symbiotic mind is not only because the quantity free energy

provides a measure of cognition across spatiotemporal scales but because the free energy principle and its various corollaries suggest a plausible criterion for identifying biological cognizers across various various spatiotemporal skills so that's why they touch the fep um just for that reason which i thought was nice i'm going to ask a nice question from the chat from stephen wrote is there a dual information geometry between intrinsic and extrinsic states of active inference and another dual information geometry between action stage estates and niche spaces can i give a first answer please okay i would say that the total state space which includes the variables for the internal the blanket and the external space that's the shape it is and then there's going to be sub shapes like you could only look at the blanket and the internal states that's the particular states so that's partitioning the system of interest away from the niche or you could look at it in a in a four way or there's probably a lot of other ways but it seems like the full model is the full geometry and then there's going to be some reduced geometries and then stephen clarified i.e a dual information geometry in relation to the generative model and what it is modeling and a dual information geometry between what actions are happening and the niche modification i think that gets at also that sigma mapping function in the decosta paper with like okay after there's conditional independence of the kind that the markov blanket partitioning provides there's still so are some internal state partitions in the world that maybe do some prediction about or anticipation about external states so it's as if they're estimating external let's talk to somebody who knows information geometry that's it i was going to say but my guess would be depends on where you put the mark on blanket but definitely i think uh anybody out there who's listening that knows about information geometry i'm curious to take a deep dive down that path so so i encourage you to get in touch and contact us come talk to us about information geometry because i would like to learn some more there was a distinction drawn in the paper between adaptive and mere active inference so i actually don't think that this has come up we've seen several distinctions like instrumentalism and realism in terms of the interpretation but this is i think another relevant distinction fep falls short of making the claim that all markov blankets draw a line around cognizers it has been suggested that what determines whether a living markov blanketed system is cognizer is whether or not it is the kind of thing which engages in adaptive active rather than mere active inference kirchoff 2018 adaptive active inference requires that a system autonomously engage in active inference maximizing sensory evidence for its own existence so there's been a coupled oscillator model as well as the uh flywheel governor model another classic kind of mechanical regulatory system as well as the discussion well does a rock have a markov

blanket that type of question and so this is from the kirchoff 2018 paper just drawing the distinction between mere active inference which is like the ability to draw a partition around a system at all to distinguish it from its niche and its interfaces and then its generative model could be like nil or boring or just a coin flip to contrast that with adaptive active inference which is actually the inactive sensory motor affordance semantic coupling i'm a strange loop type system what do you think about that blue so i wonder you know when we talk about active inference and adaptive active adaptive active inference um implies more than just mental action or maybe mental action is sufficient but but i think i feel like adaptive active inference implies that there's more of an action component and you know we've talked about situations where where you can have active inference occurring without necessarily taking an action or maybe a mental action um you can actively infer what's happening in the environment without performing some kind of action that's adaptive to it i don't know maybe that's that's that exactly it's action and non-action it can be adaptive to not act so is that mirror waiting good question right right right and also like you can have adaptation that's occurring in the brain like you adapt to a noise in the environment like it's super loud and sound super loud but then eventually your brain adapts so are you still is that mirror or is that adaptive nice so emergence do you want to read whatever or say whatever you'd like to say about immersions sure so um this is just the wikipedia definition so i'll just read it um emergence in philosophy systems theory science an art emergence occurs when an entity is observed to have properties its parts do not have on their own properties or behaviors which emerge only when the parts interact in a wider hole and i just included these references so so that that definition kind of echoes integrated information theory of consciousness which again implies that like consciousness is this emerging property when the parts are interacting and create something that's bigger than the whole and then um both of these um wait oh yeah so both of these papers um the roses paper and then this paper by varley and eric hole start to get into the mathematics of emergence and they use similar metrics it's like they use the partial information decomposition over the the five metric that's used in the integrated information theory of consciousness to kind of quantify emergence like what is this how much is this emergent property or it's what degree is this an emergent property of these parts um and then what this is a strong and weak emergence um paper that you you included daniel oh yep so i just i saw it it was really cool that first that you chose some papers from people in the active inference community and that their recent papers like within the last year or two so just shows that there's development on these topics and that's interesting to think about and then i just thought

to throw in a third like two truths and a lie but like two newer citations and one older one just the qualitative distinction between strong and weak emergence was influential by mark badao from 1997 that paper says i conclude that the scientific and philosophical prospects for weak emergence are bright and so it's easy to be like oh i want like the best version of whichever thing but in the paper it distinguishes between strong emergence as being sort of the metaphysical discussion about how wholes take on meanings greater than the sum of their parts and sort of all versus none like snapping into existence in this kind of semi-magical way versus weak emergence like the ant colony the city the actual complex systems that can be modeled and so that kind of operationalized emergence while allowing a channel of metaphysical discussion like you just mentioned consciousness and other people bring up emergent properties consciousness all the time like integrated information theory it's an consciousness so it's kind of just interesting how there's like a metaphysical thread in what this integration means as well as a uh operational and a more empirical side of this and i also saw a recent 2021 paper i can't remember the author but um dealing with fmri and emergence like in fmri which is kind of interesting i i didn't i didn't include it but um i thought that that was cool it was 20 21. okay um that's the mark of blankets go for it that's it so again this camp this comes from that uh here cough paper 2018. um and you know this this figure is called blankets all the way down um so you know here they show a single-celled organism and they predict what they they illustrate one more conflict i model the world um and then you have a multi-cell organism a blanket of blankets that says we modeled the world um and then here it shows a pontiff that shows blankets within blankets we model ourselves modeling the world and that was something uh interesting that david krakauer actually said at the summer institute um this last summer he said you know humans is one of the things that makes us uniquely human is that we build models of our model we build models of our models um so i thought that that was interesting cool and then on the right side so there's the uh ppp example protozoa plants pontiffs and then another cyber physical way that the markov blanket concept and the partitioning of systems this way is being used is in digital systems so this is like a team member and then there's a certain type of blanket that they're interfacing through in terms of sense and action and the database can also be thought of as having different kinds of sensory and action states as well as internal generative model it's dealing with a different external set of states that's from the vayatkin at all 2020 how about multi-scale integration you can go for it so i'll just read the yellow here um and so this is from the ramstead owl paper 2019 which i think that that's where this figure is from and then there's also the hess but owl 2019 paper it

says the multi-scale integrationist view is a pluralist's theory about cognitive boundaries it uses the notion of nested markov blankets to demonstrate that cognitive systems have a plurality of ontological boundaries each relevant to the study of cognition according to this view each of the spatio-temporally nested components of this cognitive system may be ontologically picked out by deploying the markov blanket formalism at different scales and that's what we were talking about earlier we talked about individuality at different scales and we talked about the markov blanket individuating systems so so this is just one way that multi-scale integration can be approached but before the markov blanket even was an idea there was other approaches to partitioning multi-scale systems so let's just go right to the author's distinction in the paper of what uni scale integration is and how it relates to the adaptive versus mirror active inference discussion and this is a question stephen also raised in the chat multi-skill integration so that's that nesting of systems within one another just not i'm intentionally not adding any adjectives just any kind of nesting systems multi-scale integration of living systems that failed to possess the high degree of autonomy required of and hence fail to be cognizers continue to engage in mere active inference at fast time scale eg a mitochondria in a cell so that is visualized here before returning to the quote like the circular nodes on the bottom let's just say are doing mere active inference they're simple sensors or actuators and then they sort of just in a very clean way lead to increasingly weakly emergent behavior that starts approximating adaptive active inference so for example you have a python script that is adaptive active inference just so we don't have to even debate whether you know snails or birds are doing it just you write a script that is adaptive active inference the sub functions within it even though they might have an input output structure and even a generative model they're going to be a lot more mechanical or simple or linear not displaying the properties of the actual active agent in the python script at the higher level so that's describing this process of uniscale integration just the paper describes more about how adaptive active inference at the slower scale of the cognizing system that's the thought bubble here drives the continued integration of the nested markov blankets at scales below it and not vice versa so the pythons variables drive the patterns of the computation of the lower level functions but it does those don't meaningfully feed back into higher level properties so adaptive active influence at the high end and then that's just being merely explained mechanistically by unidirectional integration with systems that are decreasingly cognizer-like on some continuum of cognition so i i like the the term that's used here and i think i'll highlight why when we come to um the nested cone slide but he says the behavior of constituent nested

markov blanket systems are enslaved to the slower dynamics of the autonomizing cognizing organism so i like this concept of enslaving um the sub routines cognitive evolution do you wanna say anything uh sure so this is um just i'll just read this little excerpt from the people paper because the author didn't technically use the term cognitive evolution and i thought that it might um refer to a couple different things but uh i'll read this quote because i think it's maybe highlights the the point that the author was getting at so this is a question about the kinds of physiological individuals that we can reasonably ascribe the term cognizer to it's a question about how to count biological minds the significance of this question lies in the fact that its answer may be used to shed light upon possible major transitions in cognition perhaps there are many such transitions the move from reflex behavior to sensory motor coordination the move from non-sentience to sentience to move from individual intentionality to group intentionality which this again echoes the the major transition of of evolution um so cognitive evolution i thought about it in a couple ways and we're coming to the cone slide next so maybe we'll make a flow forward you know i thought that that's one way that we can think about cognitive evolution from like the transitions in evolution from say multicellular to or single cell to multicellular um but here this is from um mike levin's paper the computational boundary of the self and you can see this is like the the cognitive um cone right so so this is the cognitive cone of an individual that the like cones that were you know inverted and used in this way but you can see that the the cognitive cone of a tip is much smaller than that of a dog is much smaller than that of a human and so perhaps this cognitive evolution is um you know in terms of of scale of of the organism or or you know cognitive capacity of an organism but mike here uses this uh idea of nested kind of this multi-scale nested cognition and here it's shown here the cell the organism the colony to represent these compound intelligences and when we talked on the stream back in july like we talked about the warping of the um the the cone that like the bigger cone but like so the colony enslaves the behavior of the organism enslaves the behavior of the cell and so this goes all the way down and that's why i liked um what uh matthew sims his term enslaving um in these nested more complicated structures um because and the idea that like if you were to grab and twist the top of this cone you would grab and twist all the cones in it depending on how tight you twisted right so so i like this idea of enslaving the behavior and this was a good visual to kind of tie that all together that terminology also arises from synergetics from hakken not fuller as they say um and this is footnote 10 an order parameter is a notion taken from synergetics and used in dynamic systems theory it denotes a measure of a global system's macro scale unstable slow dynamics

that's like the sleep wake that enslaves the fast dynamics of microscale component systems that's like the eeg rhythm and results in a globally emergent pattern so it's about multi-scale systems organizing but synergetics hackin not fuller approaches it from a dynamic systems fuller approaches it from a geometric perspective so there's other ways to do this partitioning mark how blanket is not the only partitioning out there but this was cool about also um levin's partitioning which we just got to learn about all right on to the squid so i i don't know if these are the exact right species but one of the key uh empirical examples of the paper the one that's uh followed up on do you want to describe the system sure well so i mean the vibrio fischeri um are the bacteria that colonize the squid and yes it's the hawaiian bobtail squid and shout out to michelle nishiguchi and her lab that like was the lab underneath my lab when i was in biology lab so the juvenile squid um recruits the vibrio fisheries into the what's called a light organ and that the bacteria are bioluminescent um and so it becomes essentially like inoculated with these um and so then it says i'll just read this because it's easy it's probably a better description so it's a nocturnal predator of the shallow reef um the berries and sands during the day and hunts at night so it's always whether or not it is preyed upon it's always influenced with this association with these bioluminescent bacteria so the juvenile is inoculated into the light organ by through seawater and then the host promotes the colonization of uh vibrio fisheries and only vibrio fisheries through the production of mucus um which is bacteria and food the elimination of competing bacteria through hemocyte defenses of its innate immune system and the eventual shedding of the ciliated dependent appendages and swelling of the crypt membranes preventing further entry into the light organ so the switch sucks up the vibrio fisheries says we want you we only want you all of the other bacteria we're not going to allow them like we're just not going to facilitate that interaction um and then it says following gradients gradients of chitin which they feed upon vibrio fischeri migrate deeper into the crypts of the light organized organ colonizing it and causing a biochemical reaction resulting in their emission of bioluminescent light this occurs just in time for the bobtails nightly comfort prey and something else i thought that was important for this um cycle is that they're expelled every at the every morning right like so they just get out like so so like they colonize the the vocal split colonizes and sucks up all the bacteria they multiply and divide inside of the of the squid and then it shoots them out so they're back in the sea water and they're available for recapture by other um squid we're going to return to the markov blanket partitioning consider that squid system and how using the mark out blanket partitioning we can distinguish between uni-directional and reciprocal so the nomenclature in this paper are the internal states

maybe because there's r in the word then s and a are the usual words for the state of sensory sense states coming in to r and then active action states going out of r influenced by so these are like the set of isolating nodes from which knowing them you're knowing as much as you can about r the internal states and then here ϕ or v i don't know which one it is is the external states figure two is where we uh turn to these two different kinds of multi-scale integration and directly compare them using kind of the nomenclature and the contribution i believe of this specific so to understand the difference uh i will deploy the terminology of users you and resources are users are living systems the internal states of which inferentially generate subpersonal predictions that allow them to use other markov blanketed systems that are external to them so here is like the spider making its web so it's a user of this extended cognitive resource and so we can imagine that there's a mark of blanket partitioning that's around these entities like we don't need to deny that there is a partitioning between the web and the spider we can think about it using this u and r maybe this unr is a partition we can ask the author like is it a partitioning of just markov blankets from each other is there going to be some sort of interface is that the niche or that's the unidirectional case the bi-directional case is where those two interacting systems in the same way that the spider can be thought of as the unidirectional user of the resource of the web that the two interactants are sort of using and benefiting from each other as an extended cognitive resource so that is the one of the main points of the paper looking at that difference between kind of mere multi-scale systems and then ones where it's not just like a ladder it's almost like a conversation that is giving some new emergence outcomes very cool okay the next slide just shows some more discussion on the new u and r relationships that arise and then how the new non-equilibrium steady state density arising from that interaction in the synergetic sense acts as a ordering parameter for u_1 and u_2 so the constituents are part of a new ordering regime because of the bi-directional to be clear each constituent biocognizer continues to engage in adaptive active inference acting in ways that ensure it remains statistically separate from its environment so again in the unidirectional case it's mirror active inference on the bottom that becomes increasingly cognitive as you go towards some level of autonomy like a pyramid scheme this is like a different structure where there's increasing levels of autonomy of different kinds of interacting systems and then there's sort of a bigger wrapper around that cool it's like the the interaction creates a new cone a new tent it builds a tent yep makes sense though then this slide brings this emergence from the functional because it's one thing to say that the functional um you know these two brain regions get coupled together now there's a functional difference in their

firing rates that's kind of a low mechanistic bar the higher bar to pass is this concept of ontological emergence like it really becomes a thing in and of itself the authorized that's related to these two ideas that uh first that some properties of the macro system cannot be reduced to the structural properties of the component parts and they're governing microdynamics so that's sort of anti-reductionism or holism where that exists there's a strong argument that you're kind of losing something if you keep on dissecting and then two because of this irreducible a macro level system has ontological status eg is an entity in its own right so that just is a good question to think about and to ask the author just which systems are unidirectional integrators which systems are bi-directional how do we know why does it matter those are just all good questions to ask so another question that i have is to what degree does do these new properties have to exist like if we're going to quantify in the very quantitative sense is it any number greater than zero or where are we drawing the line at what number of emergent properties or what quant what quantity of new properties creates a new system yeah it comes up with uh integrated information theory and a lot of other quantitative measures of integration and consciousness because it's like okay so if this system's one and this one's a hundred thousand are you gonna have now a second level theory like 500 is the cutoff or is it true that i could you know trade one of these for 100 000 of these like once you start putting everything on a numerical value and then that becomes your action selection metric oh if it's worth the square root of this then it's worth this so it sets up a market for action based upon these metrics which is pretty interesting that's a bi-directional emergence definitely figure three is using the markov blanket partitioning to look at the squid system so uh the squid is like user one resource two so for species one it's the squid bacteria system so resource species pair one is the squid and it's first user and second resource v fisheries u2 r1 so these show the two kinds of interfaces like they're encircling each other so it's kind of like embedded one is embedded within the other from a causal perspective they're the niche for each because even though when you put them on the two-dimensional map they're all just like flat there on the plane their nearest neighbors are either themselves or the other one so that's kind of a cool way to expand the topology of the just the histology of the tissue and then pull out like actually you know each beta cell in the pancreas is surrounded by other cells that aren't beta cells something like that and the other ones in a weird sense are also surrounded by the other kind of cell even though there's a lot more of this non-beta cell type so just a cool way to look at tissue interactions from the mark all blanket perspective you want to describe for or what or what so so here in this um figure it shows the the squid cycle um which we have

talked about uh we just we just kind of talked about it so um it shows at time zero the squid is resting after venting um so there's some vibrio fishery left inside of the squid they're not like you know sterilizing themselves and so there's this is the repopulation of um vibrio fischeri and so then after 12 hours there's a high density of um vibrio fischeri and then it shows this degree of interest so on the y-axis degree of integration is the degree of biocognitive individuality and so right here when there's the most squid inside vibrio fisheries um that's when there's the peak of integration presumably because the squid with you know three vibrio fisheries inside of it is not going to light up but when it concentrates it has a whole lot of the vibrio fischeri inside of the light organ then that's when it's luminescent um and that's when you know the novel properties so like not the you know well the you know vibration right alone wouldn't be sufficient to be um like there's not enough of them to be you know luminescent and then the split alone without the vibrofischerie doesn't light up and then it's not capable of of predation in the way that it normally does and so it's the peak of integration happens right before the sun comes up and then at dawn the vibrio fisheries are expelled and then the bioluminescence is over so that's that's kind of when there's no um integration between the two or limited or minimal integration between the two species thanks for the explanation so a few things on this it's like there's no single cut off this dashed line it's kind of like how many grains of sand in the heap like one is usually you know too few but 100 maybe you're getting there it's this continuum of cognitive embeddedness and of emergent function and there is an actual time that that is most functional and that's just because the world has an actual structure where there's a certain time where there's the most light and it's the darkest outside and then there's a time where it's ineffective so if we can discount it when it's ineffective and low density and we can say that this is like the peak of that natural variability when it's the most bright of the bioluminescence and the most dark outside there's something that's varying between those two levels and so this was just a cool way to show that that there's a smooth increase and that it's a little bit like model specific where we choose to draw the line with the with who's the biological cognizer so this was a cool pattern and to include the environmental variation in the cognition so it's not like here's when you're good at taking tests or not it was like here's the environment as it changes and then here's how the organism this reminds me of like you know building a nightlight out of fireflies so i don't know in california do you guys have fireflies there when i was a little kid they were everywhere on the east coast and so we would catch fireflies but like how many fireflies does it take to make a nightlight like you can't have one fight well it depends on the size of your jar and how many can you

catch like you can't create a nightlight with one firefly like there's just not enough light but if you get like you know 20 in there then you have a good like nightlight so true how many make a party right okay this is fun um implications of the vibrio squid so just a nice uh argument by nature approach that i'm sure we could learn more about that how do you go from an observation in the natural world oh this bird you know the one with this chromosome set does this for this long ergo this is possible or this is something about nature in general so there are three philosophical implications that are exposed by this striking association between the vibrio and the squid that are significant to the account of symbiotic minds which follows so the first one is that the vibrio squid is a clear example of a symbiotic physiological individual and that's distinguished uh from the non-physiological individual systems like collections of physiological individuals you know if i eat food the other person doesn't get the nutrients but then you go okay but one person eating does provide the nutrients for another person so it's going to be a continuum and there's all that's like sort of the complexity of it but it's a great example of integration the like i like it and then the second one is um physiological individuals or matters of degree over time the same living system can be more or less of a symbiotic physiological individual so i think that really takes that um almost unsaid criticism of the first point and immediately builds on it to say like yes it is a continuum that's the whole interesting part you can see how the system's cognitive interfaces change through time and that's the interesting story like for humans certainly our ability to um do something alone in a room is gonna change through time when we're born it's very low and then there's an age range where you can do some task alone in a room and then eventually there isn't so it's like that is something that varies continuously just like there's the daily rhythm there's a lively rhythm to some cognitive elements of all so it's not reducing them or explaining them away it's actually looking at that variability and what it says about the system the last philosophical implication may be thrown into relief when asking the critical question what is it that produces the bioluminescence the squid the bacteria or the temporary assemblage of these organisms is that a thing that deserves a noun of itself because we have nouns for assemblages of other things of one type like the barrel of oil or something like that but is there a unit and a noun or are we left to describe this only with a and then default implicitly back to the darwinian replicator oh the squid's just trying to reproduce and the bacteria is doing competition so they're the individuals ergo is a tenuous relationship or it's an interra interaction ecologically but it's not a thing in and of itself because it's not the replicator those are great questions raised by the author so the questions that i maybe have for the author um the bioluminescence is there in the vibrio

fisheries it's just self-detectable by us or presumably by the prey that the vibrio fisheries hunting right so so i think it's again like it's a continuum it's not uh um there is bioluminescence or there's not bioluminescence there's like more or less bioluminescence or detectable bioluminescence that occurs at you know x lumens so so what is the the line there that is wrong good question so here's one of the last points in the discussion today it's again integrating cognitive and functional and physiological individuality together and then um what is in the background of you why are those when we say individuality and we have to qualify it by saying well i mean informational or i mean physiological what are we qualifying it implicitly or explicitly against um i think the materialist worldview might suggest that the individual is like just the the one corpuscular entity like the one thing that can be separated out like the ant is the nest mate worker now there's an argument for the ampy and the colony and the nest made being a tissue but people say oh i saw 10 ants on the windowsill not like i saw a thousandth of an ants on the windowsill so there's i think a pure materialist argument for just separability and that having the more noun-like characteristic the author is arguing that um there's this folk biological concept that sometimes prevents thinking about physiological individuals as real uh bona fide and that's this like evolutionary individualism model uh which can even have a collaborative or optimistic bent to it but it still draws lines around what levels of individuality exist in the world because they're the ones that most cleanly resemble darwinian replicators so not the systems that grow slowly outwards or transfer through different media in the same way and then the argument is that fep and its corollary active inference the mirror and adaptive and the continuum between like we talked about offer an ideal program for bringing our folk psychological intuitions in line with the way nature has carved its own joints over evolutionary time skills so that's an interesting claim definitely last slide this was like a william blake line that i thought of when i saw the uni-directional the bird a nest the spider a web man friendship people friendship so it's a great description of kind of this modification and i think it's getting at this idea that there's niche modification in nature and we do it that's sort of the unambiguous part you know we change our environment but then we're also our own environment so then there's some dynamics of like culture and communication and being each other's niche that are very different than the the bacteria changing the ph of a solution it can go up it can go down but it's still ph but then there's this different boundedness to what can happen with humans working together so that just brought a close these few questions like where is the multi-scale integration happening how how do we identify the unidirectional components and the reciprocal components and uh great paper so

just a final thought and something that i've thought about a lot um in terms of of reading this paper because it's this colonization of the squid by this bacteria i think a lot about us and our colonization i mean there are more non-human cells in the body than there are human cells which is fascinating so but we also have the ability to manipulate our niche our internal niche the niche of the microbes that reside within us through taking antibiotics but there are lots of other things probiotics and not just antibiotics but but just even what you eat and where you go and how whether you're breastfed or not or go through vaginal birth or not and all these things contribute to your own inoculation or how you inoculate your children and this is something that you can do like literally you can change the composition of your microbiota like through your lifetime is that niche modification is it what is that exactly i think it's one reason why a um not neutral or unbiased but flexible model like being presented here we can just say okay rather than wondering whether it falls on one side or the other of a binary classification like yes it's co um symbiotic or not we just have a descriptive framework first and then you can think about the implications later or separate them somehow but we hope that if you're listening in the right time you'll join us for the 30.1 and 0.2 because the author hopefully will join for at least the first one and it's cool topics and we can do a little bit like we did in 29.2 let's think of some systems are they unidirectional are they reciprocal what reciprocal systems could exist like what does that mean so sounds good yep thanks uh people who are listening and steven for the questions live thanks blue for collaborating on the slides here and thanks to the author too for providing us such such great food for thought totally agreed so thanks and see you later